external transform space

$$V_{24} = \left(\frac{4\pi}{5!}\right)\left(\frac{4\pi}{!5}\right)\left(\frac{4\pi}{35}\right)^1\left(\frac{4\pi}{18}\right)^2\left(\frac{4\pi}{32}\right)^3\left(\frac{4\pi}{8}\right)^4$$

$$\rho_{\text{Leech}} = \left(\frac{4\pi}{5!}\right)\left(\frac{4\pi}{!5}\right)\left(\frac{4\pi}{35}\right)^1\left(\frac{4\pi}{18}\right)^2\left(\frac{4\pi}{32}\right)^3\left(\frac{4\pi}{8}\right)^4$$

$5\left(\frac{4\pi}{2}\right)\left(\cos\left(\frac{7}{5}\right)\right)^2 = \frac{8}{e^{\phi_4}}$ mass

$2\left(\frac{5}{\sqrt{7}}\right)^2\left(\cos\left(\frac{5}{2}i\right)\right)^2\left(\cos\left(\frac{7}{5}\right)\right)^2 = \frac{32}{e^{\phi_3}}$ temperature

$\left(\frac{5}{\sqrt{7}}\right)\left(\frac{3^{-1/3}}{W_{We}}\right) = \frac{18}{e^{\phi_2}}$ charge

$\left(\sinh\left(\sinh\left(\frac{1}{7}\right)\right)\right)^{-1} = \frac{35}{e^{\phi_1}}$ length

$\left(\frac{4\pi}{4}\right)\left(\sinh\left(\frac{1}{4}\right)\right)^2 = \frac{!5}{e^{\phi_0}}$ time

$\phi_4 = 2.1764268381757\ldots$ $\quad 2(3-1)^2 = 8$
$\phi_3 = 1.4167869859079\ldots$ $\quad 2(3+1)^2 = 32$
$\phi_2 = 1.8755459671396\ldots$ $\quad 2(3\pm 0)^2 = 18$
$\phi_1 = 1.6162591817564\ldots$ $\quad 7\times 5 = 35$ $\quad \Gamma(5) = 24$
$\phi_0 = 5.3912583683231\ldots$ $\quad !5 = 44$ $\quad 5! = 120$

$m_p = 2.1764268381757\ldots\times 10^{-8}$ kg $\quad m_p = \phi_4\times 10^{-8}$ kg
$T_p = 1.4167869859079\ldots\times 10^{32}$ K $\quad T_p = \phi_3\times 10^{32}$ K
$q_p = 1.8755459671396\ldots\times 10^{-18}$ C $\quad q_p = \phi_2\times 10^{-18}$ C
$l_p = 1.6162591817564\ldots\times 10^{-35}$ m $\quad l_p = \phi_1\times 10^{-35}$ m
$t_p = 5.3912583683231\ldots\times 10^{-44}$ s $\quad t_p = \phi_0\times 10^{-44}$ s

$$A_{\text{ext}}\,B_{\text{ext}}\,(1 + A_{\text{int}}\boxtimes) \qquad \boxtimes = \frac{l_p m_p}{{q_p}^2}\times\frac{\text{coulomb}^2}{\text{meter kilogram}}$$

$$\frac{1}{x} + x + \frac{x^3}{2\pi} = (i^i)^{-\frac{4\pi}{8}} - \frac{m_p}{\text{kilogram}}$$

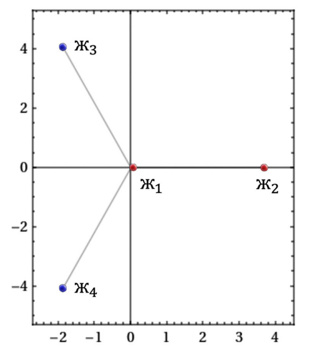
$ж_1 = 0.0854245431533304\ldots$
$ж_2 = 3.66756753485501\ldots$
$ж_3 = -1.87649603900417\ldots + 4.06615262615972\ldots\, i$
$ж_4 = -1.87649603900417\ldots - 4.06615262615972\ldots\, i$

$ж_r = 4.47826244916751\ldots$ $\quad ж_3 = ж_r\, e^{ж_\theta i}$
$ж_\theta = 2.00316562310924\ldots$ $\quad ж_4 = ж_r\, e^{-ж_\theta i}$

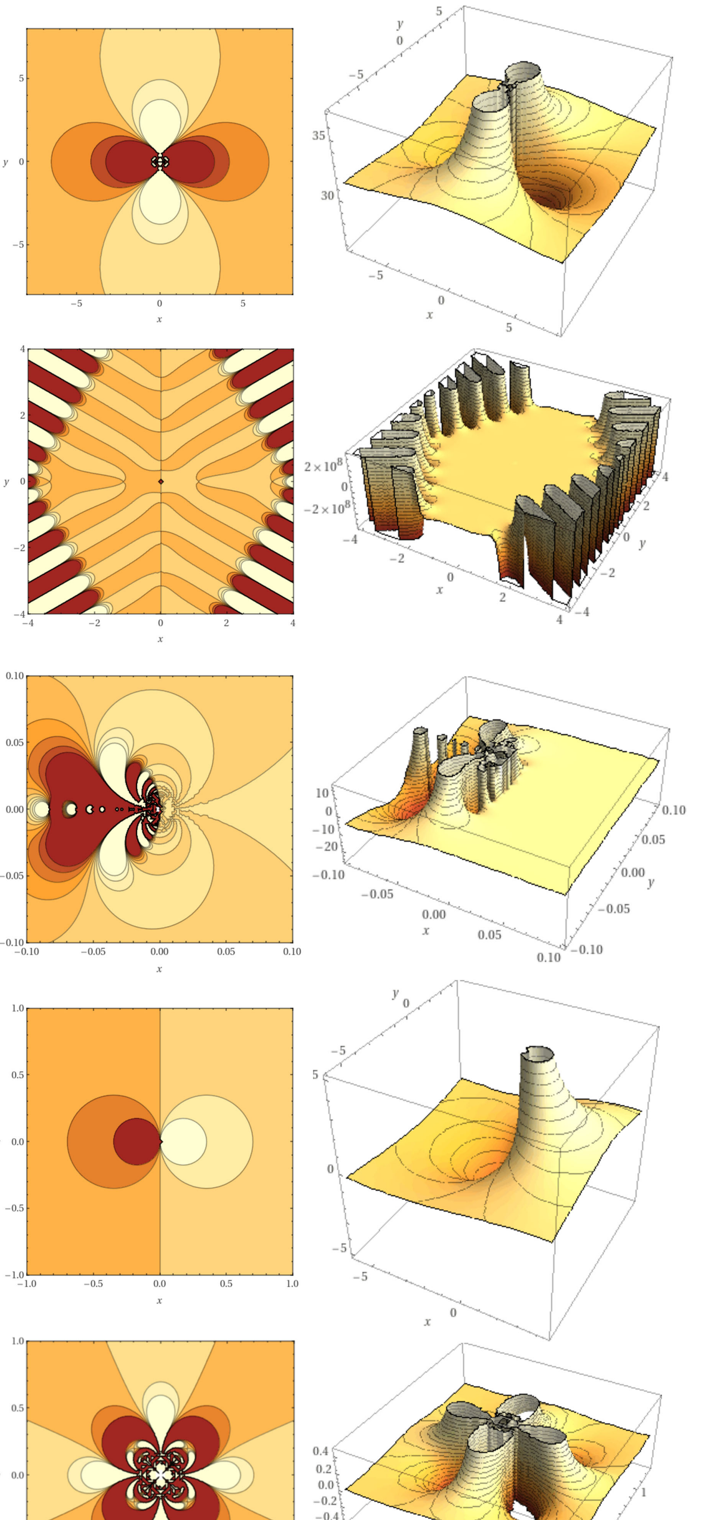
$ж_1ж_2ж_3ж_4 = 2\pi$ $\quad -Im(ж_3) = Im(ж_4)$
$ж_1 + ж_2 + ж_3 + ж_4 = 0$ $\quad Im(ж_1) = 0$
$-ж_1 - ж_2 - ж_3 - ж_4 = 0$ $\quad Im(ж_2) = 0$
${ж_1}^2 + {ж_2}^2 + {ж_3}^2 + {ж_4}^2 = -4\pi$
$-{ж_1}^2 - {ж_2}^2 - {ж_3}^2 - {ж_4}^2 = 4\pi$ $\quad \frac{ж_1 + ж_2}{ж_3 + ж_4} = -1$
$ж_1 + ж_2 = -2\,Re(ж_3) = -2\,Re(ж_4)$
$ж_1 - ж_2 = Re(ж_1) - Re(ж_2)$ $\quad \frac{{ж_1}^2{ж_2}^2}{2\pi} = \frac{2\pi}{{ж_3}^2{ж_4}^2}$
$ж_3 + ж_4 = 2\,Re(ж_3) = 2\,Re(ж_4)$
$ж_3 - ж_4 = 2i\,Im(ж_3) = -2i\,Im(ж_4)$
$Re(ж_1) + Re(ж_2) + Re(ж_3) + Re(ж_4) = 0$
$Im(ж_1) + Im(ж_2) + Im(ж_3) + Im(ж_4) = 0$
$Re(ж_3)^2 + Re(ж_4)^2 = \frac{1}{2}(ж_1 + ж_2)^2$
$Im(ж_3)^2 + Im(ж_4)^2 = \frac{1}{2}(ж_3 - ж_4)^2$
$ж_1ж_2 + ж_3ж_4 + ((ж_1 + ж_2)(ж_3 + ж_4)) = 2\pi$
$ж_1ж_2 + ж_1ж_3 + ж_1ж_4 + ж_2ж_3 + ж_2ж_4 + ж_3ж_4 = 2\pi$
$Re(ж_3)\,Re(ж_4) - Im(ж_3)\,Im(ж_4) = {ж_r}^2$
$Re(ж_3)^2 + Im(ж_3)^2 = {ж_r}^2$
$Re(ж_3)^2 + Im(ж_4)^2 = {ж_r}^2$ $\quad ж_3ж_4 = {ж_r}^2$
$Re(ж_4)^2 + Im(ж_3)^2 = {ж_r}^2$ $\quad ж_1ж_2 = \frac{2\pi}{{ж_r}^2}$
$Re(ж_4)^2 + Im(ж_4)^2 = {ж_r}^2$

$(a^2 + b^2)(c^2 + d^2) = (ac - bd)^2 + (ad + bc)^2 = {ж_r}^4$
$(a^2 + b^2)(c^2 + d^2) = (ac + bd)^2 + (ad - bc)^2 = {ж_r}^4$

$a = Re(ж_3)$ $\quad b = Im(ж_3)$ $\quad c = Re(ж_4)$ $\quad d = Im(ж_4)$
$a = Re(ж_4)$ $\quad b = Im(ж_4)$ $\quad c = Re(ж_3)$ $\quad d = Im(ж_3)$
$a = Im(ж_3)$ $\quad b = Re(ж_3)$ $\quad c = Im(ж_4)$ $\quad d = Re(ж_4)$
$a = Im(ж_4)$ $\quad b = Re(ж_4)$ $\quad c = Im(ж_3)$ $\quad d = Re(ж_3)$



$\sum_{n=1}^{4} ж_n = 0$
$\prod_{n=1}^{4} ж_n = 2\pi$
$\sum_{n=1}^{4} {ж_n}^2 = -4\pi$



$\log((i^i)^0) = ж_1 + ж_2 + ж_3 + ж_4$
$\log((i^i)^4) = -ж_1ж_2ж_3ж_4$
$\log((i^i)^8) = {ж_1}^2 + {ж_2}^2 + {ж_3}^2 + {ж_4}^2$

$S_{n-1}(r) = \frac{2\pi^{n/2}}{\Gamma\left(\frac{n}{2}\right)} r^{n-1}$
$V_n(r) = \frac{\pi^{n/2}}{\Gamma\left(\frac{n}{2}+1\right)} r^n$

$\log((i^i)^4) = -1\,ж_1ж_2ж_3ж_4$
$\log((i^i)^8) = -2\,ж_1ж_2ж_3ж_4$
$\log((i^i)^{12}) = -3\,ж_1ж_2ж_3ж_4$
$\log((i^i)^{16}) = -4\,ж_1ж_2ж_3ж_4$

$\frac{S_{n+1}(r)}{V_n(r)} = 2\pi r$
$\frac{S_{n-1}(r)}{V_n(r)} = \frac{n}{r}$

$\log((i^i)^8) = 1({ж_1}^2 + {ж_2}^2 + {ж_3}^2 + {ж_4}^2)$
$\log((i^i)^{16}) = 2({ж_1}^2 + {ж_2}^2 + {ж_3}^2 + {ж_4}^2)$
$\log((i^i)^{24}) = 3({ж_1}^2 + {ж_2}^2 + {ж_3}^2 + {ж_4}^2)$
$\log((i^i)^{32}) = 4({ж_1}^2 + {ж_2}^2 + {ж_3}^2 + {ж_4}^2)$

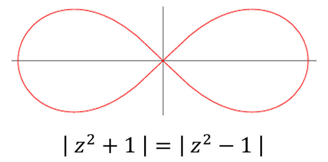
$(i^i)^{-1} = \frac{s^4\,{W_{We}}^4}{8}$
$(i^i)^{-2} = e^{\pi}$
$(i^i)^{-4} = e^{2\pi}$
$(i^i)^{-6} = e^{3\pi}$
$(i^i)^{-8} = e^{4\pi}$

$\sum_{n=0}^{\infty}\frac{1}{n!} = \frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \frac{1}{3!} + \frac{1}{4!} + \frac{1}{5!} + \frac{1}{6!} + \frac{1}{7!} + \frac{1}{8!} + \cdots = e$
$\sum_{n=0}^{\infty}\frac{(-1)^n}{n!} = \frac{1}{0!} - \frac{1}{1!} + \frac{1}{2!} - \frac{1}{3!} + \frac{1}{4!} - \frac{1}{5!} + \frac{1}{6!} - \frac{1}{7!} + \frac{1}{8!} - \cdots = \frac{1}{e}$
$\sum_{n=0}^{\infty} x^n = x^0 + x^1 + x^2 + x^3 + x^4 + x^5 + x^6 + x^7 + \cdots = \frac{1}{1-x}$
$\sum_{n=0}^{\infty} (-x)^n = x^0 - x^1 + x^2 - x^3 + x^4 - x^5 + x^6 - x^7 + \cdots = \frac{1}{1+x}$

$\left[Li_2\left(\frac{1}{\varphi_i}\right) + Li_2(\varphi_i)\right] = \frac{(4\pi)^2}{288}$
$i\left[Li_2\left(\frac{1}{\varphi_i}\right) - Li_2(\varphi_i)\right] = V_{fe}$
$i\left[Li_2\left(\frac{1}{{\varphi_i}^2}\right) - Li_2({\varphi_i}^2)\right] = \frac{2V_{fe}}{3}$
$i\left[Li_2\left(\frac{1}{{\varphi_i}^3}\right) - Li_2({\varphi_i}^3)\right] = 0$
$i\left[Li_2\left(\frac{1}{{\varphi_i}^4}\right) - Li_2({\varphi_i}^4)\right] = -\frac{2V_{fe}}{3}$
$i\left[Li_2\left(\frac{1}{{\varphi_i}^5}\right) - Li_2({\varphi_i}^5)\right] = -V_{fe}$
$i\left[Li_2\left(\frac{1}{{\varphi_i}^6}\right) - Li_2({\varphi_i}^6)\right] = 0$
$i\left[Li_2\left(\frac{1}{i}\right) - Li_2(i)\right] = 2K$
$i\left[Li_2\left(\frac{1}{i^2}\right) - Li_2(i^2)\right] = 0$
$i\left[Li_2\left(\frac{1}{i^3}\right) - Li_2(i^3)\right] = -2K$
$i\left[Li_2\left(\frac{1}{i^4}\right) - Li_2(i^4)\right] = 0$
$\left[Li_2\left(\frac{1}{i}\right) + Li_2(i)\right] = -\frac{1}{6}\left(\frac{4\pi}{8}\right)^2$
$Li_2(1) - Li_2(-1) = \left(\frac{4\pi}{8}\right)^2$
$Li_2(-1) = -\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{8}\right)$
$Li_2(1) = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{4}\right)$
$\sum_{n=1}^{\infty}\frac{1}{n^2} = Li_2(1)$
$\sum_{n=1}^{\infty}\frac{(-1)^n}{n^2} = Li_2(-1)$
$\sum_{n=1}^{\infty}\frac{1}{(2n+1)^2} = \left(\frac{4\pi}{16}\right)\left(\frac{4\pi}{8}\right)$
$\sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)^2} = K$
$\sum_{n=1}^{\infty}\frac{\sin\left(n\frac{\pi}{2}\right)}{n^2} = K$
$\sum_{n=1}^{\infty}\frac{\sin\left(n\frac{\pi}{3}\right)}{n^2} = G_{Gi}$
$\sum_{n=1}^{\infty}\frac{\cos\left(n\frac{\pi}{3}\right)}{n^2} = \left(\frac{4\pi}{\Gamma(5)}\right)^2$
$\sum_{n=1}^{\infty}\frac{\cos\left(n\frac{\pi}{2}\right)}{n^2} = -\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{32}\right)$
$\sum_{n=0}^{\infty}\binom{n}{k} = 2^n$
$\sum_{n=1}^{\infty}\frac{1}{n^s} = \zeta(s)$ $\quad \sum_{n=0}^{\infty}\frac{H_n}{n^2} = 2\,\zeta(3)$
$\sum_{n=0}^{\infty}\varphi^{-n} = \varphi^2$ $\quad \sum_{n=0}^{\infty}\psi^{-n} = \psi^3$
$\sum_{n=1}^{\infty}\frac{F_n}{F_{2n}} = \mathcal{L}_{RL}$
$\sum_{n=1}^{\infty}\frac{1}{F_n} = \mathcal{F} = \sqrt{5}\sum_{n=1}^{\infty}\frac{1}{\varphi^n - (-\varphi)^n}$
$\sum_{n=1}^{\infty}\frac{1}{L_n} = \mathcal{L}_{RL} = \sum_{n=1}^{\infty}\frac{1}{(-\varphi)^{-n} + \varphi^n}$
$\sum_{n=0}^{\infty}\frac{x^n}{n!} = \frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \cdots = e^x$

lemniscate

$(x^2 + y^2)^2 = (x^2 - y^2)$



$|z^2 + 1| = |z^2 - 1|$

defines points equidistant from roots 0, 1, and −1

$Li_2(\varphi_i) = \left(\frac{4\pi}{\Gamma(5)}\right)^2 + G_{Gi}\, i$
$Li_2\left(\frac{1}{\varphi_i}\right) = \left(\frac{4\pi}{\Gamma(5)}\right)^2 - G_{Gi}\, i$
$Li_2(i) = -\left(\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{32}\right)\right) + K\, i$
$Li_2\left(\frac{1}{i}\right) = -\left(\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{32}\right)\right) - K\, i$
$\log((i^i)^{-\pi/2}) = \left(\frac{4\pi}{8}\right)^2$
$\log((i^i)^{-32\pi}) = (4\pi)^2$
$\log((i^i)^8) = -4\pi$
$\log((i^i)^4) = -2\pi$
$\log((i^i)^2) = -\left(\frac{4\pi}{4}\right)$
$\log((i^i)^{1/1}) = -\left(\frac{4\pi}{8}\right)$
$\log((i^i)^{1/2}) = -\left(\frac{4\pi}{16}\right)$
$\log((i^i)^{1/3}) = -\left(\frac{4\pi}{\Gamma(5)}\right)$
$\log((i^i)^{1/4}) = -\left(\frac{4\pi}{32}\right)$
$\frac{\mathcal{S}}{2} = \sin\left(\frac{3\pi}{14}\right) + 1$
$\frac{\mathcal{S}}{2} = \cos\left(\frac{4\pi}{14}\right) + 1$
$e^{\pi i} = -1$
$e^{\pi} = (i^{-i})^2$
$i^{-i} = \frac{s^4\,{W_{We}}^4}{8}$
$e^{\pi} = \frac{s^8\,{W_{We}}^8}{8^2}$
$i^{-i} = 2\,L^4\,{W_{We}}^4$
$e^{\pi} = 4\,L^8\,{W_{We}}^8$
$(i^i)^{-4\pi/8} = \frac{s^{2\pi}\,{W_{We}}^{2\pi}}{\sqrt{8^{\pi}}}$
$(i^i)^{-2\pi} = \frac{s^{8\pi}\,{W_{We}}^{8\pi}}{8^{2\pi}}$
$W_{We} = \frac{\sqrt{\sqrt{8}}}{s}\, e^{\left(\frac{4\pi}{32}\right)}$
$W_{We} = \frac{1}{L}\left(\frac{i^{-i}}{2}\right)^{1/4}$
$W_{We} = \frac{1}{L}\left(\frac{e^{\pi}}{4}\right)^{1/8}$
$W_{We} = \frac{1}{L}\left(\frac{e^{2\pi}}{16}\right)^{1/16}$
$W_{We} = \frac{1}{L}\left(\frac{e^{4\pi}}{8\cdot 32}\right)^{1/32}$
$e^{\zeta'(0)} = \frac{1}{\sqrt{2\pi}}$
$e^{\beta'(0)} = \frac{s}{\sqrt{4\pi}}$
$\omega_1 = \frac{\Gamma^3\left(\frac{1}{3}\right)}{4\pi}\,\varphi_i$
$\omega_2 = \frac{\Gamma^3\left(\frac{1}{3}\right)}{4\pi}$
$s = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{2\pi}}$
$L = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{8\pi}}$
$L_1 = \frac{\Gamma^2\left(\frac{1}{4}\right)}{\sqrt{32\pi}}$
$L_2 = \frac{2\pi}{\sqrt{2}}\frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma^2\left(\frac{1}{4}\right)}$
$C_U = 2\pi\frac{\Gamma\left(\frac{1}{2}\right)}{\Gamma^2\left(\frac{1}{4}\right)}$
$G_{Ga} = \frac{\Gamma^2\left(\frac{1}{4}\right)}{2\pi\sqrt{2\pi}}$

$\frac{r_n}{r_e} = \left(\frac{4\pi}{\Gamma(5)}\right)^2$
$\frac{r_n}{r_+} = K$
$G_{Gi} = V_{\text{ideal tetrahedron}}$
$2G_{Gi} = V_{fe}$
$2K = 2\pi\, C_d$
$4K = 2\pi\log(D_d)$
$4K = V_{\text{Whitehead link}}$
$4K = V_{\text{ideal octahedron}}$
$8K = V_{\text{Borromean rings}}$
$\sin\left(\frac{3\pi}{14}\right) = \cos\left(\frac{4\pi}{14}\right)$
$\sin\left(\frac{4\pi}{14}\right) = \cos\left(\frac{3\pi}{14}\right)$
$\left(\frac{4\pi}{8}\right) = L\,L_2$
$\left(\frac{4\pi}{16}\right) = L_1\,L_2$ $\quad \cos\left(\frac{4\pi}{8}\right) = 0$
$\left(\frac{4\pi}{32}\right) = \frac{L\,L_1\,L_2}{s}$ $\quad \sin\left(\frac{4\pi}{8}\right) = 1$
$\int_0^{\pi} {\mathrm{Cl}_2}^2(x)\,dx = \zeta(4)$
$\int_0^{2\pi/12} -4\log|e^{-xi} - e^{xi}| = V_{fe}$
$\int_0^{2\pi/6} -6\log|e^{-xi} - e^{xi}| = V_{fe}$
$\int_0^{1/2}\Gamma(1+x)\,\Gamma(1-x)\,dx = \frac{4K}{2\pi}$
$\int_0^{1}\Gamma(1+x)\,\Gamma(1-x)\,dx = \frac{8K}{2\pi}$
$\int_0^1\int_0^1\frac{1}{1+x^2y^2} = K$
$\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\frac{1}{2}\,2\pi}$
$\int_{-\infty}^{\infty} e^{-x^4}\,dx = \sqrt{\frac{L}{2}\sqrt{2\pi}}$
$\int_{-1}^{1}\sqrt{1+x^2}\,dx = P_{up}$
$\int_{-1}^{1}\sqrt{1-x^2}\,dx = \left(\frac{4\pi}{8}\right)$
$\int_{-\infty}^{\infty}\frac{1}{1+x^2}\,dx = \pi$
$\int_{-\infty}^{\infty}\frac{1}{1+x^4}\,dx = \frac{4\pi}{\sqrt{32}}$
$\int_0^1\frac{1}{\sqrt{1+x^4}}\,dx = \frac{s}{\sqrt{32}}$
$\int_0^1\frac{1}{\sqrt{1-x^4}}\,dx = \frac{s}{4}$
$\int_0^1\sqrt{1-x^4}\,dx = \frac{s}{6}$
$\int_0^1\frac{x^2}{\sqrt{1-x^4}}\,dx = \frac{\pi}{s}$
$\int_0^{1/2}\int_0^{x} 6\sqrt{x^2+y^2}\,dy\,dx = \frac{P_{up}}{8}$
$\int_0^{\infty} 2^{-x}\,dx = \frac{1}{\log(2)}$
$\int_0^{\infty}\frac{\log(x)}{e^x}\,dx = \gamma$
$\int_0^{1}\log\left(\log\left(\frac{1}{x}\right)\right)dx = \gamma$

Combinatorics

$V_{\text{unit sphere}} = \left(\frac{4\pi}{3}\right)$
$V_{\text{Gabriel's horn}} = \left(\frac{4\pi}{4}\right)$
$V_{\text{pesudosphere}} = \left(\frac{4\pi}{6}\right)$
$V_{\text{exp trumpet}} = \left(\frac{4\pi}{8}\right)$
$V_{\text{cos spindle}} = \left(\frac{4\pi}{4}\right)\left(\frac{4\pi}{8}\right)$

$A_{\text{one pedal}} = 1$
$A_{\text{unit lemniscate}} = 2$
$A_{\text{exp trumpet}} = \pi\, P_{up}$
$A_{\text{cos spindle}} = 2\pi\, P_{up}$
$A_{\text{unit circle}} = 2\pi$
$A_{\text{unit sphere}} = 4\pi$
$A_{\text{pesudosphere}} = 4\pi$
$A_{\text{unit cardioid}} = 6\pi$

$L_{\text{unit wave}} = \frac{4\pi}{s} + s$
$L_{\text{unit cardioid}} = 8$
$L_{\text{unit cardioid cusp normal}} = \sqrt{\frac{32}{18}}$

$\alpha_F = \lim_{n\to\infty}\frac{d_n}{d_{n+1}}$
$\delta_F = \lim_{n\to\infty}\frac{a_{n-1} - a_{n-2}}{a_n - a_{n-1}}$

3D sphere packing densities

$\rho_{\text{cubic lat}} = \left(\frac{4\pi}{\Gamma(5)}\right)$
$\rho_{\text{hex lat}} = \frac{2}{\sqrt{3}}\left(\frac{4\pi}{\Gamma(5)}\right)$
$\rho_{\text{tet lat}} = \frac{\sqrt{3}}{2}\left(\frac{4\pi}{32}\right)$
$\rho_{\text{BCC}} = \sqrt{3}\left(\frac{4\pi}{32}\right)$
$\rho_{\min} = \frac{1}{\sqrt{2}}\left(\frac{4\pi}{18}\right)$
$\rho_{\text{FCC}} = \frac{4\pi}{\sqrt{288}}$
$\rho_{\text{HCP}} = \frac{4\pi}{\sqrt{288}}$

densest hypersphere lattice packings

$\delta_2 = \sqrt{3}\left(\frac{4\pi}{\Gamma(5)}\right)$
$\delta_3 = \sqrt{2}\left(\frac{4\pi}{\Gamma(5)}\right)$
$\delta_4 = \left(\frac{4\pi}{16}\right)^2$
$\delta_5 = \sqrt{2}\left(\frac{4\pi}{5!}\right)\left(\frac{4\pi}{4}\right)$
$\delta_6 = \sqrt{3}\left(\frac{4\pi}{16}\right)\left(\frac{4\pi}{18}\right)\left(\frac{4\pi}{32}\right)$
$\delta_7 = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{35}\right)\left(\frac{4\pi}{8}\right)$
$\delta_8 = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{16}\right)^3$

max density sphere packings

$\rho_{2D} = \sqrt{\left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{8}\right)}$
$\rho_{3D} = \frac{4\pi}{\sqrt{288}}$
$\rho_{4D} = \left(\frac{4\pi}{16}\right)^2$
$\rho_{8D} = \left(\frac{4\pi}{\Gamma(5)}\right)\left(\frac{4\pi}{16}\right)^3$

fixed point equation for $i^{i^{i^{\cdot^{\cdot}}}}$

$x = i^x$
$x = \cos\left(\frac{4\pi}{8}x\right) + i\sin\left(\frac{4\pi}{8}x\right)$
$\log(x) = \left(\frac{4\pi}{8}\right)xi$
$x = e^{\left(\frac{4\pi}{8}\right)xi}$

factors of 288

1	288
2	144
3	96
4	72
6	48
8	**36**
9	**32**
12	24
16	**18**

$\Gamma(\varphi + 1) = \Gamma(\varphi - 1)$
$\Gamma(\varphi_i + 1) = -\Gamma(\varphi_i - 1)$
$\Gamma\left(-\frac{1}{\varphi} + 1\right) = \Gamma\left(-\frac{1}{\varphi} - 1\right)$
$\Gamma\left(\frac{1}{\varphi_i} + 1\right) = -\Gamma\left(\frac{1}{\varphi_i} - 1\right)$

fixed point equation for $\pm C_{CFP}$

$e^x(x - 1) = e^{-x}(x + 1)$
$x = \frac{e^x + e^{-x}}{e^x - e^{-x}}$
$\coth(x) = x$
$\operatorname{csch}(C_{CFP}) = L_{LL}$
$\operatorname{csch}(-C_{CFP}) = -L_{LL}$
$\sinh(C_{CFP}) = \frac{1}{L_{LL}}$
$\sinh(-C_{CFP}) = -\frac{1}{L_{LL}}$

$\zeta\left(\tan\left(\frac{4\pi}{32}\right) + 1\right) = \zeta\left(\cot\left(\frac{4\pi}{32}\right) - 1\right)$
$\zeta\left(-\tan\left(\frac{4\pi}{32}\right) - 1\right) = \zeta\left(-\cot\left(\frac{4\pi}{32}\right) + 1\right)$

$\prod_{n=1}^{\infty}\left(1 - \left(-\frac{1}{\varphi^2}\right)^n\right) = F_{FF}$
$\prod_{p\text{ prime}}\left(\frac{p^2+1}{p^2-1}\right) = \frac{5}{2}$
$\prod_{p\text{ prime}}\left(\frac{p^4+1}{p^4-1}\right) = \frac{7}{6}$
$\prod_{p\text{ prime}}\left(\frac{p^s+1}{p^s-1}\right) = \frac{\zeta(s)^2}{\zeta(2s)}$
$\prod_{p\text{ prime}}\left(1 + \frac{1}{p^s}\right) = \frac{\zeta(s)}{\zeta(2s)}$
$\prod_{p\text{ prime}}\left(1 - \frac{1}{p^s}\right) = \frac{1}{\zeta(s)}$
$\prod_{p\text{ prime}}\left(1 + \frac{1}{p(p-1)}\right) = \left(\frac{35}{4\pi}\right)\left(\frac{18}{4\pi}\right)\left(\frac{8}{4\pi}\right)^2\zeta(3)$

$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$
$\Gamma\left(-\frac{1}{2}\right) = -\sqrt{4\pi}$
$\Gamma\left(\frac{1}{2}\right)\Gamma\left(-\frac{1}{2}\right) = -2\pi$
$\Gamma\left(\frac{3}{2}\right) = \sqrt{\frac{4\pi}{16}}$
$B\left(\frac{1}{2},\frac{3}{2}\right) = \left(\frac{4\pi}{8}\right)$
$B\left(\frac{1}{2},\frac{1}{2}\right) = \pi$
$B\left(\frac{1}{2},\frac{1}{4}\right) = s$

fixed point equation for D_{Do}

$\cos(x) = x$
$x = Re(e^{xi})$

fixed point equation for Ω

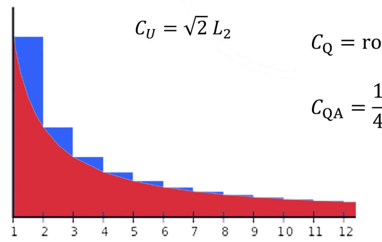
$x = e^{-x}$
$x = -\log(x)$
$x = \log\left(\frac{1}{x}\right)$

$(1 + i)!\,(1 - i)! = \frac{4\pi}{e^{\pi} - e^{-\pi}}$
$(0 + i)!\,(0 - i)! = \frac{2\pi}{e^{\pi} - e^{-\pi}}$
$|\Gamma(0 + i)|^2 = \frac{2\pi}{e^{\pi} - e^{-\pi}}$
$\left|\Gamma\left(\frac{1}{2} + i\right)\right|^2 = \frac{2\pi}{e^{\pi} + e^{-\pi}}$
$|\Gamma(1 + i)|^2 = \frac{2\pi}{e^{\pi} - e^{-\pi}}$

$\lim_{n\to\infty}\frac{1}{\log(p_n)}\prod_{k=1}^{n}\frac{1}{1 - \frac{1}{p_k}} = e^{\gamma}$
$\lim_{n\to\infty}\log(p_n)\prod_{k=1}^{n}\frac{1}{1 + \frac{1}{p_k}} = \frac{6}{e^{\gamma}}\left(\frac{4\pi}{\Gamma(5)}\right)^2$
$\lim_{x\to 0}\frac{1}{x}\left(\frac{1}{\Psi(1-x)} - \frac{1}{\Psi(1+x)}\right) = \frac{6}{\gamma^2}\frac{(4\pi)^2}{288}$
$\lim_{n\to\infty}\frac{\pi^0}{\Gamma(1)} + \frac{\pi^1}{\Gamma(2)} + \frac{\pi^2}{\Gamma(3)} + \cdots + \frac{\pi^n}{\Gamma(n+1)} = e^{\pi}$
$\lim_{x\to 0}\frac{1}{x}\left(\frac{1}{\Gamma(1+x)} - \frac{1}{\Gamma(1-x)}\right) = 2\gamma$
$\lim_{x\to 0}\left(\frac{\zeta(1+x) + \zeta(1-x)}{2}\right) = \gamma$
$\lim_{n\to\infty}\left(\sum_{k=1}^{n}\frac{1}{k} - \int_1^n\frac{1}{x}\,dx\right) = \gamma$
$\Gamma'(1) = -\gamma$

$s = 2L$
$L = 2L_1$
$L_1 = \frac{s}{4}$
$L_2 = \frac{\pi}{s}$
$G_{Ga} = \frac{s}{2\pi}$
$C_U = \sqrt{2}\,L_2$

$s = 4\,K(i)$
$C_d = \frac{2K}{2\pi}$
$D_d = e^{\left(\frac{4K}{2\pi}\right)}$
$D_d = e^{2C_d}$
$\mathcal{S} = \log((e^{\gamma}\,C_U)^2)$
$C_Q = \text{root of }\zeta\left(\frac{1}{2},\frac{x}{2}\right)$
$C_{QA} = \frac{1}{4}\zeta\left(\frac{3}{2},\frac{C_Q}{2}\right)$



$\varphi = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \ddots}}}}$

$\frac{1}{\varphi} = 0 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{1 + \ddots}}}}$

$C_{CF} = 0 + \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{1}{3 + \cfrac{1}{4 + \cfrac{1}{5 + \ddots}}}}}$

$C_{R1} = 0 + \cfrac{e^{-0\pi}}{1 + \cfrac{e^{-2\pi}}{1 + \cfrac{e^{-4\pi}}{1 + \cfrac{e^{-6\pi}}{1 + \cfrac{e^{-8\pi}}{1 + \ddots}}}}}$

$\frac{1}{1 - e} = 0 + \cfrac{1}{1 + \cfrac{2}{2 + \cfrac{3}{3 + \cfrac{4}{4 + \cfrac{5}{5 + \ddots}}}}}$

$e = 2 + \cfrac{1}{1 + \cfrac{1}{2 + \cfrac{2}{3 + \cfrac{3}{4 + \cfrac{4}{5 + \ddots}}}}}$

$\left(\frac{4\pi}{8}\right) = 1 + \cfrac{1}{1 + \cfrac{1}{1 + \cfrac{1}{3 + \cfrac{1}{1 + \cfrac{1}{4 + \cfrac{1}{5 + \ddots}}}}}}$

$\left(\frac{4\pi}{16}\right) = 0 + \cfrac{1}{1 + \cfrac{1}{3 + \cfrac{1}{5 + \cfrac{1}{7 + \cfrac{1}{9 + \ddots}}}}}$

$\left(\frac{16}{4\pi}\right) = 1 + \cfrac{1^2}{2 + \cfrac{3^2}{2 + \cfrac{5^2}{2 + \cfrac{7^2}{2 + \cfrac{9^2}{2 + \ddots}}}}}$

$\left(\frac{s^2}{4\pi}\right) = 2 + \cfrac{1^2}{4 + \cfrac{3^2}{4 + \cfrac{5^2}{4 + \cfrac{7^2}{4 + \cfrac{9^2}{4 + \ddots}}}}}$

$\left(\frac{4\pi}{4}\right) = 3 + \cfrac{1^2}{6 + \cfrac{3^2}{6 + \cfrac{5^2}{6 + \cfrac{7^2}{6 + \cfrac{9^2}{6 + \ddots}}}}}$

$18\left(\frac{2\pi}{s^2}\right) = 4 + \cfrac{1^2}{8 + \cfrac{3^2}{8 + \cfrac{5^2}{8 + \cfrac{7^2}{8 + \cfrac{9^2}{8 + \ddots}}}}}$

$K = 0 + \cfrac{1^4}{1 + \cfrac{1^4}{8 + \cfrac{3^4}{16 + \cfrac{5^4}{24 + \cfrac{7^4}{32 + \ddots}}}}}$

$2K = 1 + \cfrac{1\cdot 1}{3 + \cfrac{1\cdot 2}{1 + \cfrac{2\cdot 2}{3 + \cfrac{2\cdot 3}{1 + \cfrac{3\cdot 3}{3 + \ddots}}}}}$

$\frac{\pi}{2} = 1 + \cfrac{1\cdot 1}{1 + \cfrac{1\cdot 2}{1 + \cfrac{2\cdot 3}{1 + \cfrac{3\cdot 4}{1 + \cfrac{4\cdot 5}{1 + \ddots}}}}}$

$\frac{L}{2} = 1 + \cfrac{1\cdot 1}{2 + \cfrac{2\cdot 3}{2 + \cfrac{4\cdot 5}{2 + \cfrac{6\cdot 7}{2 + \cfrac{8\cdot 9}{2 + \ddots}}}}}$

$\cosh(x) = \frac{1}{2}(e^x + e^{-x})$
$\sinh(x) = \frac{1}{2}(e^x - e^{-x})$
$\cos(x) = \frac{1}{2}(e^{xi} + e^{-xi})$
$\sin(x) = \frac{i}{2}(e^{xi} - e^{-xi})$
$\cosh(x) = \frac{x^0}{0!} + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \frac{x^8}{8!} + \cdots$
$\sinh(x) = \frac{x^1}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \frac{x^9}{9!} + \cdots$
$\cos(x) = \frac{x^0}{0!} - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \cdots$
$\sin(x) = \frac{x^1}{1!} - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \cdots$
$\tan^{-1}(x) = \frac{x^1}{1} - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \frac{x^9}{9} - \cdots$
$\log(1 + x) = \frac{x^1}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \cdots$
$\log(1 - x) = -\frac{x^1}{1} - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} - \cdots$
$\log(x) = \frac{x^{-1}}{1} - \frac{x^{-2}}{2} + \frac{x^{-3}}{3} - \frac{x^{-4}}{4} + \frac{x^{-5}}{5} - \cdots$
$Li_s(x) = \frac{x^1}{1^s} + \frac{x^2}{2^s} + \frac{x^3}{3^s} + \frac{x^4}{4^s} + \frac{x^5}{5^s} + \cdots$
$\zeta(1) = \frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \cdots = \infty$
$\zeta(2) = \frac{1}{1^2} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \cdots = \frac{\pi^2}{6}$
$\zeta(3) = \frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \cdots$
$\zeta(3) = \frac{1}{2}\int_0^1\frac{\log(x)\log(1-x)}{x(1-x)}\,dx$
$\zeta(3) = \frac{1}{2}\int_0^{\infty}\frac{x^2}{e^x - 1}\,dx$ $\quad \sqrt{\frac{\zeta(3)}{\zeta'(-2)}} = 2\pi\, i$
$\zeta(3) = \frac{2}{3}\int_0^{\infty}\frac{x^2}{e^x + 1}\,dx$ $\quad \frac{\zeta\left(\frac{3}{2}\right)}{\zeta\left(-\frac{1}{2}\right)} = -4\pi$

Pascal's triangle

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1
1 7 21 35 35 21 7 1

$\sqrt{288} = 16 + \cfrac{1}{1 + \cfrac{1}{32 + \cfrac{1}{1 + \cfrac{1}{32 + \ddots}}}}$

natural numbers $\binom{n+0}{1} = \frac{n}{1!}$
triangular numbers $\binom{n+1}{2} = \frac{n(n+1)}{2!}$
tetrahedral numbers $\binom{n+2}{3} = \frac{n(n+1)(n+2)}{3!}$
pentatope numbers $\binom{n+3}{4} = \frac{n(n+1)(n+2)(n+3)}{4!}$
5 − simplex numbers $\binom{n+4}{5} = \frac{n(n+1)(n+2)(n+3)(n+4)}{5!}$

$1^2 + 2^2 + 3^2 + 4^2 + 5^2 + \cdots + n^2 = \frac{n(n-1)(2n+1)}{3!}$
$1^3 + 2^3 + 3^3 + 4^3 + 5^3 + \cdots + n^3 = (1 + 2 + 3 + 4 + 5 + \cdots + n)^2$
$(1 + x)^n = \frac{n^0x^0}{0!} + \frac{n^1x^1}{1!} + \frac{n^2x^2}{2!} + \frac{n^3x^3}{3!} + \frac{n^4x^4}{4!} + \cdots + \frac{n^nx^n}{n!}$
$2^n = \frac{n^0}{0!} + \frac{n^1}{1!} + \frac{n^2}{2!} + \frac{n^3}{3!} + \frac{n^4}{4!} + \frac{n^5}{5!} + \frac{n^6}{6!} + \frac{n^7}{7!} + \frac{n^8}{8!} + \frac{n^9}{9!} + \frac{n^{10}}{10!} + \cdots$
$2^n = \frac{1}{0!} + \frac{n}{1!} + \frac{n(n-1)}{2!} + \frac{n(n-1)(n-2)}{3!} + \cdots$
$\frac{\pi}{4} = \frac{1}{1} - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} + \frac{1}{13} - \frac{1}{15} + \frac{1}{17} - \frac{1}{19} + \frac{1}{21} - \frac{1}{23} + \frac{1}{25}\cdots$
$\infty = \frac{2}{1}\frac{4}{3}\frac{6}{5}\frac{8}{7}\frac{10}{9}\frac{12}{11}\frac{14}{13}\frac{16}{15}\frac{18}{17}\frac{20}{19}\frac{22}{21}\frac{24}{23}\frac{26}{25}\frac{28}{27}\frac{30}{29}\frac{32}{31}\frac{34}{33}\frac{36}{35}\cdots$
$0 = \frac{2}{3}\frac{4}{5}\frac{6}{7}\frac{8}{9}\frac{10}{11}\frac{12}{13}\frac{14}{15}\frac{16}{17}\frac{18}{19}\frac{20}{21}\frac{22}{23}\frac{24}{25}\frac{26}{27}\frac{28}{29}\frac{30}{31}\frac{32}{33}\frac{34}{35}\frac{36}{37}\cdots$
$\frac{\pi}{2} = \prod_{n=1}^{\infty}\left(1 + \frac{1}{n}\right)^{(-1)^{n+1}} = \prod_{n=1}^{\infty}\left(\frac{2n}{2n-1}\cdot\frac{2n}{2n+1}\right) = \left(\frac{2}{1}\cdot\frac{2}{3}\right)\left(\frac{4}{3}\cdot\frac{4}{5}\right)\cdots$
$\frac{L}{2} = \prod_{n=1}^{\infty}\left(1 + \frac{1}{2n}\right)^{(-1)^{n+1}} = \prod_{n=1}^{\infty}\left(\frac{4n}{4n-2}\cdot\frac{4n}{4n+1}\right) = \left(\frac{2}{3}\cdot\frac{4}{5}\right)\cdots\left(\frac{7}{6}\cdot\frac{8}{9}\right)\cdots$

$n! = \int_0^{\infty}(x-1)^n\,e^{-x}\,dx$
$!n = n!\sum_{k=0}^{n}\frac{(-1)^k}{k!}$
$!0 = 1$
$!1 = 0$
$\Gamma(s) = \int_0^{\infty}\frac{x^{s-1}}{e^x}\,dx$
$Q(1,x) = e^{-x}$
$P(1,x) = 1 - e^{-x}$
$n! = \Gamma(n+1)$
$\Gamma(n) = (n-1)!$
$\frac{1}{\Gamma(-n)} = 0$
$\zeta(-2n) = 0$
$\zeta(s) = Li_s(1)$
$\zeta(0) = -\frac{1}{2}\log(e)$
$\zeta'(0) = -\frac{1}{2}\log(2\pi)$
$\zeta(s) = \frac{1}{\Gamma(s)}\int_0^{\infty}\frac{x^{s-1}}{e^x - 1}\,dx$
$\zeta(s) = \prod_{p\text{ prime}}\frac{1}{1 - p^{-s}}$
$\log(ab) = \log(a) + \log(b)$
$\log(n) = \int_1^n\frac{1}{x}\,dx$
$\frac{d}{dx}\log(x) = \frac{1}{x}$
$\log(x) = -\log\left(\frac{1}{x}\right)$
$Li_1(x) = -\log(1 - x)$
$Li_0(x) = \frac{x}{1 - x}$
$Li_{-1}(x) = \frac{x}{(1 - x)^2}$
$Li_{-2}(x) = \frac{x(1 + x)}{(1 - x)^3}$

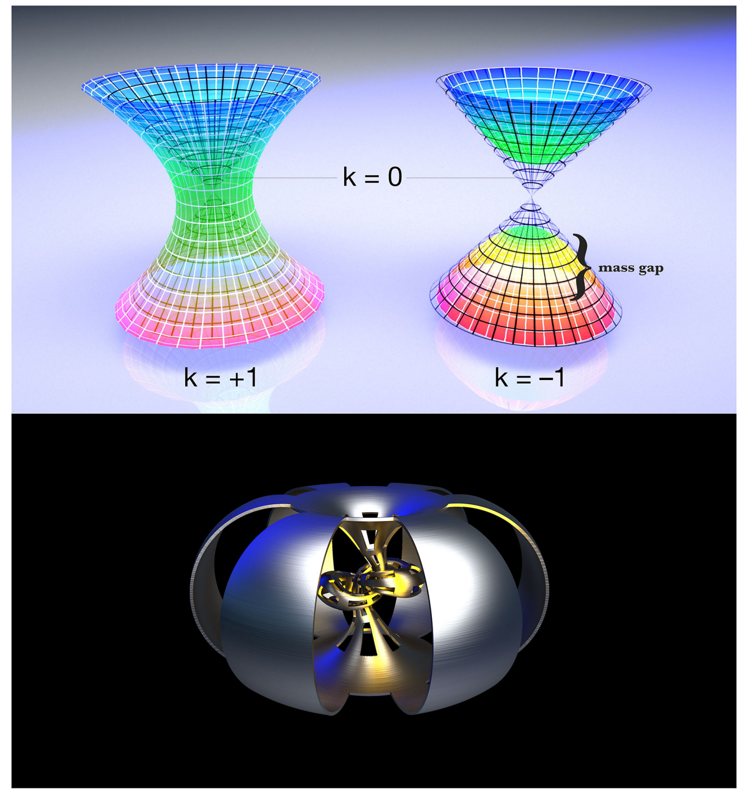
$\binom{n}{k} = \frac{n!}{k!\,(n-k)!}$
$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} = \prod_{k=1}^{n}\frac{n+k}{k}$
$(a + b)^n = \sum_{k=0}^{n}\binom{n}{k}a^kb^{n-k}$
$|i| = 1$ $\quad P + 1 = P^3$
$|\varphi_i| = 1$ $\quad P - 1 = P^{-4}$
$i^4 = 1$ $\quad \varphi = \frac{a}{b} = \frac{a+b}{a}$
$\varphi_i^6 = 1$ $\quad \varphi_i = \frac{a}{b} = \frac{a-b}{a}$
$i = (-1)^{1/2}$ $\quad \varphi + 1 = \varphi^2$
$\varphi_i = (-1)^{1/3}$ $\quad \varphi - 1 = \frac{1}{\varphi}$
$i = e^{\pi i/2}$ $\quad \varphi_i - 1 = -\frac{1}{\varphi_i}$
$\varphi_i = e^{\pi i/3}$
$\varphi = e^{\pi i/5} + e^{-\pi i/5}$
$\sqrt{5} = \frac{1}{\varphi} + \varphi$
$F(n) = \frac{\varphi^n - (-\varphi)^{-n}}{\sqrt{5}}$
$\varphi = \lim_{n\to\infty}\frac{F_{n+1}}{F_n}$
$\varphi^n = \varphi^{n-1} + \varphi^{n-2}$
$\varphi_i^n = \varphi_i^{n-1} - \varphi_i^{n-2}$
$\varphi_i = \frac{\omega_1}{\omega_2}$
$Re(\omega_1) = \frac{1}{2}\,\omega_2$
$Im(\omega_1) = \frac{\sqrt{3}}{2}\,\omega_2$

$\varphi = \sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1 + \cdots}}}}$
$\varphi_i = \sqrt{-1 + \sqrt{-1 + \sqrt{-1 + \sqrt{-1 + \cdots}}}}$
$P = \sqrt[3]{1 + \sqrt[3]{1 + \sqrt[3]{1 + \sqrt[3]{1 + \cdots}}}}$
$\varphi = \sqrt{1^2 + \sqrt{1^2 + \sqrt{1^2 + \sqrt{1^2 + \cdots}}}}$
$\varphi_i = \sqrt{i^2 + \sqrt{i^2 + \sqrt{i^2 + \sqrt{i^2 + \cdots}}}}$

$\sin(x)$ = sine
$\cos(x)$ = cosine
$\tan(x)$ = tangent
$\cot(x)$ = cotangent
$\sinh(x)$ = hyperbolic sine
$\cosh(x)$ = hyperbolic cosine
$\coth(x)$ = hyperbolic cotangent
$\operatorname{csch}(x)$ = hyperbolic cosecant
$\log(x)$ = natural logarithm
$Li_n(x)$ = polylogarithm of order n
$Q(a,x)$ = Q-regularized gamma function
$P(a,x)$ = P-regularized gamma function
$\Psi(x)$ = digamma function
$\Gamma(x)$ = gamma function
$n!$ = factorial
$!n$ = derangement
$\zeta(x)$ = Reimann zeta function
$B(x)$ = beta function
$\mathrm{Cl}_s(x)$ = Clausen function
$K(i)$ = complete elliptic integral, 1st kind
$Im(x)$ = imaginary part of x
$Re(x)$ = real part of x
$\prod$ = product
$\sum$ = sum
$\int$ = integral
r_+ = proton radius
r_n = neutron radius

Physics Monastery
Thad Roberts
October 26, 2025

$\mathrm{Cl}_2(x) = \sum_{n=1}^{\infty}\frac{\sin(nx)}{n^2} = \frac{\sin(1x)}{1^2} + \frac{\sin(2x)}{2^2} + \frac{\sin(3x)}{3^2} + \frac{\sin(4x)}{4^2} + \frac{\sin(5x)}{5^2} + \cdots$
$\mathrm{Cl}_2\left(2\pi n \pm \frac{\pi}{3}\right) = \pm G_{Gi}$ $\quad \mathrm{Cl}_2\left(2\pi n \pm \frac{2\pi}{3}\right) = \pm\frac{2G_{Gi}}{3}$ $\quad \frac{d}{dx}\mathrm{Cl}_2(x) = \mathrm{Cl}_1(x)$
$\mathrm{Cl}_2(n\pi) = 0$ $\quad \max = G_{Gi}$ $\quad \min = -G_{Gi}$ $\quad \mathrm{Cl}_2(-x) = -\mathrm{Cl}_2(x)$



legend

$t_p = 5.39125836832313\ldots\times 10^{-44}$ s — Planck time
$l_p = 1.61625918175645\ldots\times 10^{-35}$ m — Planck length
$q_p = 1.87554596713962\ldots\times 10^{-18}$ C — Planck charge
$T_p = 1.41678698590795\ldots\times 10^{32}$ K — Planck temperature
$m_p = 2.17642683817579\ldots\times 10^{-8}$ kg — Planck mass

5 — number of dimensions
7 — break in scale symmetry

$ж_1 = 0.0854245431533304\ldots$ — 1st hyperbolic partition constant
$ж_2 = 3.66756753485501\ldots$ — 2nd hyperbolic partition constant
$ж_3 = -1.87649603900417\ldots + 4.06615262615972\ldots\, i$ — 3rd hpc
$ж_4 = -1.87649603900417\ldots - 4.06615262615972\ldots\, i$ — 4th hpc
$ж_r = 4.47826244916751\ldots$ — hyperbolic radius constant
$ж_\theta = 2.00316562310924\ldots$ — hyperbolic radian constant

$G_{Gi} = 1.01494160640965\ldots$ — Gieseking's constant
$V_{fe} = 2.02988321281930\ldots$ — figure eight knot hyperbolic volume
$K = 0.9159655941772190\ldots$ — Catalan's constant
$C_d = 0.291560904030818\ldots$ — domino tiling constant
$D_d = 1.79162281206959\ldots$ — dimer constant
$\omega_1 = 0.764977018528596\ldots + 1.32497062714087\ldots\, i$ — omega_1 constant
$\omega_2 = 1.529954037057192\ldots$ — omega_2 constant

$i^{i^{i^{\cdot^{\cdot}}}} = 0.438282936727032\ldots + 0.360592471871385\ldots\, i$ — power tower of i
$i = (-1)^{1/2}$ — imaginary unit
$\varphi_i = (-1)^{1/3}$ — imaginary golden ratio
$\varphi = 1.61803398874989\ldots$ — golden ratio
$\psi = 1.46557123187676\ldots$ — supergolden ratio
$\pi = 3.14159265358979\ldots$ — Archimedes' constant
$P_{up} = 2.29558714939263\ldots$ — universal parabolic constant
$s = 5.24411510858423\ldots$ — arc length of the unit lemniscate
$L = 2.622057554292119\ldots$ — lemniscate constant
$L_1 = 1.31102877714605\ldots$ — 1st lemniscate constant
$L_2 = 0.599070117367796\ldots$ — 2nd lemniscate constant
$C_U = 0.847213084793979\ldots$ — ubiquitous constant
$G_{Ga} = 0.834626841674073\ldots$ — Gauss's constant

$C_{CF} = 0.697774657964819\ldots$ — continued fraction constant
$C_{R1} = 0.998136044598509\ldots$ — Ramanujan's 1st continued fraction constant
$M = -1.74756459463318\ldots$ — Madelung constant

$e = 2.71828182845904\ldots$ — Euler's number
$\gamma = 0.577215664901532\ldots$ — Euler-Mascheroni constant
$\Omega = 0.567143290409783\ldots$ — omega constant
$S = 0.822825249678847\ldots$ — Sierpiński's constant
$j_{0,1} = 2.404825557695772\ldots$ — 1st root of the Bessel function
$C_{CFP} = 1.19967864025773\ldots$ — Real fixed point of the hyperbolic cotangent
$D_{Do} = 0.739085133215160\ldots$ — Dottie number
$L_{LL} = 0.662743419349181\ldots$ — Laplace limit

$\alpha_F = 2.50290787509589\ldots$ — alpha Feigenbaum constant
$\delta_F = 4.66920160910299\ldots$ — delta Feigenbaum constant
$\mathcal{S} = 3.24697960371714\ldots$ — silver constant
$P = 1.32471795724474\ldots$ — plastic constant
$C_Q = 0.605443657196732\ldots$ — QRS constant
$C_{QA} = 2.03816937970215\ldots$ — associated QRS constant

with:
Angela Arvizu, Eschelon Azha, Kevin Sirios
Vanessa Moon, Avi Rubin, Mike Ritter
Matthew Fox, Wayne Eskridge, Jerry Gardner, WolframAlpha
David Heggli, Dan Whittaker, Shauna Montgomery, NIST/CODATA